SRS

$Ax + By + C = 0$

$\rightarrow y = mx + b$

$m = -\left(\frac{A}{B}\right)$ Coef $[_\ _\ ]\ [_]$

$c = -\left(\frac{C}{B}\right)$

→ Perceptron Loss function / Hinge Loss

→ Problem with Perceptron Trick

- we can't find the best value of w_1, w_2, b because it only classify the misclassified point.
  it don't give the best line.
- problem in Converge.
- we can't quantify our result.

→ Loss function for Perceptron

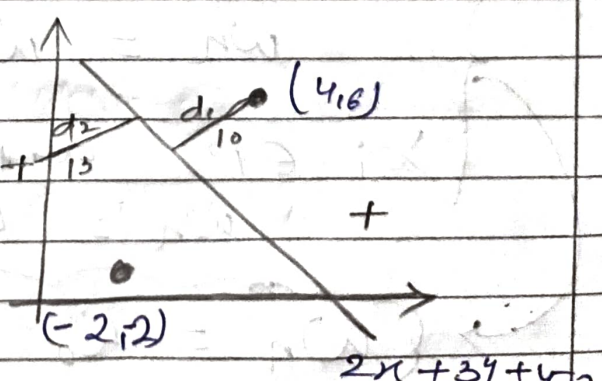
it is a fn of w_1, w_2, b.

- no. of ~~mis~~ Miss classified points

$f(w_1, w_2, b)$

$te = 10 + 13$

$\Rightarrow 2(4) + 3(6) + 4 \Rightarrow 30$



$2(-4) + 3(-2) + 4 =$

$-4 - 6 + 4 \Rightarrow -6$

$\Rightarrow |-6| = 6$

total error $= 30 + 6 \Rightarrow 36$

SRS

$$L(w_1, w_2, b) = \frac{1}{n} \sum_{i=1}^{n} L(y_i, f(x_i)) + \alpha R(w_1, w_2)$$

Regularization

$$L(y_i, f(x_i)) = \max(0, -y_i \cdot f(x_i))$$

$$f(x_i) = w_1 x_1 + w_2 x_2 + b$$

$$L = \frac{1}{n} \sum_{i=1}^{n} \max(0, -y_i f(x_i))$$

→ Minimum

n = no. of rows in data

Gradient Descent

→ Gradient Descent

$$L = \underset{w_1, w_2, b}{\operatorname{argmin}} \frac{1}{n} \sum_{i=1}^{n} \max(0, -y_i \cdot f(x_i))$$

where,

$$f(x_i) = w_1 x_{i1} + w_2 x_{i2} + b$$

for i in epochs:

$\eta = 0.01$

$$w_1 = w_1 + \eta \frac{\partial L}{\partial w_1}$$

$$w_2 = w_2 + \eta \frac{\partial L}{\partial w_2}$$

$$b = b + \eta \frac{\partial L}{\partial b}$$

$$\left[\frac{\partial L}{\partial w_1}, \frac{\partial L}{\partial w_2}, \frac{\partial L}{\partial b} \right]$$

$$\frac{\partial L}{\partial w_1} = \frac{\partial L}{\partial f(x_i)} \times \frac{\partial f(x_i)}{\partial w_1}$$

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SRS

$$\frac{\partial L}{\partial f(x_i)} = \begin{cases} 0 & \text{if } y_i f(x_i) \geq 0 \\ -y_i & \text{if } y_i f(x_i) < 0 \end{cases}$$

$$\frac{\partial f(x_i)}{\partial w_i} = x_{i1}$$

$$\frac{\partial L}{\partial w_1} = \begin{cases} 0 & \text{if } y_i f(x_i) \geq 0 \\ -y_i x_{i1} & " \quad " \quad < 0 \end{cases}$$

$$\frac{\partial L}{\partial w_2} = \begin{cases} 0 & \text{if } y_i f(x_i) \geq 0 \\ -y_i x_{i2} & " \quad " \quad < 0 \end{cases}$$

$$\frac{\partial L}{\partial b} = \begin{cases} 0 & " \quad " \geq 0 \\ -y_i & " \quad " < 0 \end{cases}$$

→ More Loss function.

↳ Binary Cross Entropy

$$L = -y_i \log \hat{y}_i + (1-y_i) \log 1-\hat{y}_i$$

Activation fn used sigmoid instead of summation

$$\sigma(z) = \frac{1}{1+e^{-z}}$$

→ Multiclass

Activation fn = Softmax → $f = \frac{e^{z_i}}{\sum_{j=1}^{K} e^{z_j}}$

Loss fn : categorical cross entropy

$$L = \sum_{j=1}^{M} y_j \log(\hat{y}_j)$$

Softmax Regression

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SdS LR

→ ^ Regression

Activation → Linear $z \geq 0$ 1, $z < 0$ 1

Loss fn → MSE $(y_i - \hat{y}_i)^2$

Loss fn	Activation fn	output
Hinge Loss	step	perceptron → binary classifier -1, 1
Log - Loss binary class entropy	Sigmoid	Logistic Regression 0 - 1
Categorical class entropy	Softmax	Softmax Regression Multiclass classification probability o/p
MSE	linear	Linear Regression o/p → no any

→ perceptron is a Mathematical Model that is very flexible we can use this Model for Multiple purpose.

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